

Process Characterization Master Report

A-Mab Drug Substance — PCMR-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; CEX, cation exchange chromatography; Cpk, process capability index; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance;

ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; RSD, relative standard deviation; RSM, response surface methodology; SEC, size exclusion chromatography; SOP, standard operating procedure; TFF, tangential flow filtration; TMP, transmembrane pressure; UF/DF, ultrafiltration and diafiltration; UV, ultraviolet; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

The A-Mab drug substance process is characterized and its control strategy is defined. The 8 unit operations of the drug substance train, Steps 3-10, were each studied under an approved plan and reported separately, and the work was executed on qualified scale-down models at the sending site under the master plan PCMP-001 and the pre-characterization risk assessment RA-001. The campaign supports the technology transfer planned in PTP-001, and this report consolidates its results. It follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012), and it takes the process model, the quality attribute framework and the criticality framework from the A-Mab case study (CMC Biotech Working Group 2009), which is also the source of the risk methodology RA-001 applies.

The campaign carried 37 process parameters into study, of which 21 are linked to a critical quality attribute and are held inside the design space. Of those, 1 parameter is classified as a critical process parameter and 20 as well-controlled critical process parameters. The remaining 16 are key or general process parameters, and they are controlled and monitored because a consistent process is part of the control strategy, not because any of them was shown to move a quality attribute. The single critical process parameter is the pH of the low-pH viral inactivation hold (Section 4).

All 10 drug substance quality attributes met their acceptance criteria. Capability was estimated by simulating 2,000 commercial batches, with every parameter varying about its set-point on a spread set by its normal operating range, and no simulated batch produced a value outside a criterion (Section 3). The lowest capability index in the set is 1.51, for cumulative MVM clearance, whose simulated mean lies 1.78 $\log_{10}$ above the required floor. The tightest of the attributes the process itself forms is high mannose at 1.94, with 2.99 percentage points between its simulated mean and the nearer of its two limits.

Cumulative viral clearance exceeds its requirement for both model viruses. The claim rests on 3 steps with independent mechanisms of removal or inactivation, and gives 18.87 $\log_{10}$ for XMuLV against a requirement of 16.70, and 10.03 $\log_{10}$ for MVM against 8.60. No clearance is claimed for the Protein A step or for the cation exchange step, so the cumulative figures are conservative to the extent of whatever those two steps deliver.

The campaign recorded 17 deviations. Each was investigated and dispositioned in the report for its step, and one of them invalidated a designed experiment. The first anion exchange execution ran on a load that was not representative of routine material, so the affected screening and response surface designs were re-executed in full on

requalified load (PCR-008). Of the rest, 15 were retained with a bounded impact argument and one was corrected by modelling and verification runs.

Two bounds apply to everything reported here. Capability, proven acceptable ranges and viral clearance are estimated from qualified scale-down models with the parameters held close to their set-points, so the edges of the design space have not been run at commercial scale and nothing in this report describes the behaviour of the process there. Stage 2 has not been executed, so this report presents no process performance qualification data and makes no claim about the performance of any batch made at commercial scale.

1 Introduction

1.1 The campaign and its documents

The A-Mab drug substance process was characterized in a single campaign covering Steps 3–10, in which each unit operation was studied over ranges wider than its routine control range so that the effect of every parameter on product quality and on process performance could be measured and bounded. The scope of the work was fixed before execution by RA-001, which assigned every parameter to a multivariate study, to a univariate assessment, or to no study at all, and the common methods, the acceptance framework and the criticality framework were fixed by PCMP-001. This report consolidates the results and states the outcome against each requirement the drug substance has to meet.

The package is listed in Table 1.1, and three documents sit above the per-step work. PTP-001 defines the transfer from the sending site to the receiving site and is the reason the campaign exists, RA-001 sets the study scope, and PCMP-001 is the umbrella plan over the 8 characterization plans. Each unit operation then carries a plan and a report, numbered by process step. This report rolls those reports up. It does not repeat their analyses, and where a figure here comes from one of them, that report is named.

Table 1.1: The A-Mab drug substance document package.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)

Document ID	Document class	Subject
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)

1.2 Scope of the characterization

37 process parameters were carried into study, of which 22 were studied in multivariate designs and 15 were assessed one at a time, and Table 1.2 gives the split by step together with the plan and report that cover it. Of the 8 steps, 6 carried a designed experiment with a screening stage followed by a response surface stage. Harvest and clarification (Step 4) and the ultrafiltration and diafiltration step (Step 10) carried none, because neither step forms a product quality attribute, and both were characterized univariately with reports that are correspondingly shorter.

Table 1.2: Characterization scope by unit operation, with the covering plan and report.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest and Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small-Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration	3	0	3	PCP-010 / PCR-010

The division of labour between the two design stages is the same at every step that has one. The screening design identifies which parameters have an effect on which response, and the response surface model is the predictive model, which makes it the basis of the design space and of the proven acceptable ranges reported in each unit operation report. Nothing in this report rests on a screening fit.

1.3 Regulatory and scientific basis

The campaign follows the lifecycle approach to process validation, in which Stage 1 establishes the process design and the commercial process is then qualified in Stage 2 (U.S. Food and Drug Administration 2011). The enhanced development approach of ICH Q8 was used throughout, so that prior knowledge and risk assessment identify what matters, characterization bounds it, and the result is a control strategy (International Council for Harmonisation 2009). Criticality is treated as a continuum and not as a binary label, following ICH Q9 (International Council for Harmonisation 2023b). The classification decisions follow the development and manufacturing principles of ICH Q11 (International Council for Harmonisation 2012). Viral safety is evaluated under ICH Q5A, which is the basis of the modular clearance claim in Section 6, and the transfer to the receiving site is managed under the pharmaceutical quality system of ICH Q10 (International Council for Harmonisation 2008) and planned in PTP-001.

2 Process description and performance

2.1 The process train

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch cell culture and purified on a platform train. Culture in the production bioreactor is followed by harvest and clarification, Protein A capture, a low-pH viral inactivation hold, cation exchange polishing, anion exchange polishing in flow-through mode, small-virus retentive filtration, and ultrafiltration and diafiltration into the formulation buffer. Table 2.1 lists the steps with the role each has in the control strategy, and Figure 2.1 shows the same train with the principal quality impact of each step.

Table 2.1: The A-Mab drug substance process train and the role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

A-Mab drug-substance process train

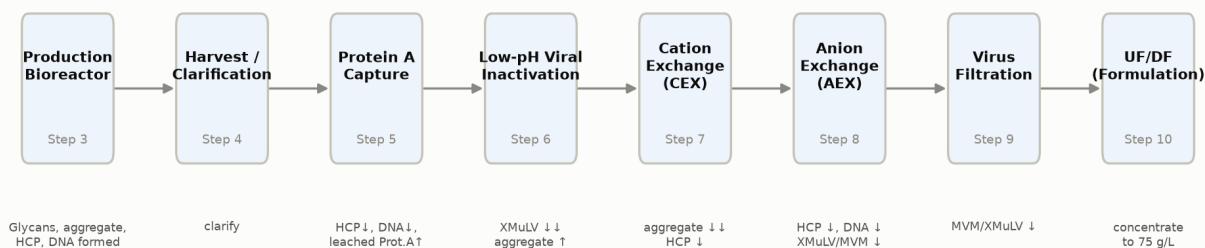


Figure 2.1: The drug substance process train, with the principal quality impact of each unit operation.

The quality burden is not spread evenly across the train. The production bioreactor sets 7 of the 10 quality attributes, which is more than any other step (Table 3.1). Cation exchange is the principal aggregate polishing step, and anion exchange and virus filtration together carry the small-virus clearance claim. The low-pH hold sets the enveloped-virus clearance claim (Section 6) and carries the only critical process parameter in the process (Section 4). The remaining steps are recovery, capture and formulation operations, and their parameters act on yield, on the impurity burden or on the physical condition of the pool. Capture is also the step at which leached Protein A enters the pool (Table 3.1).

2.2 Nominal batch performance

The nominal batch is the run at set-point that each report describes, and it is the reference point for the ranges studied around it. Culture at 15,000 L delivers a harvest titre of 4.37 g/L at a final viability of 67.1%, with a peak viable cell concentration of 13.5 e6 cells/mL. The drug substance is delivered at 75 g/L, and a batch yields 54.6 kg at the end of the train.

Product yield across the train is given in Table 2.2 and plotted in Figure 2.2. The cumulative yield is 83.2% and the lowest single-step yield is 95.2% at the cation exchange step. Yield is a process performance attribute and not a quality attribute, so no step yield is an acceptance criterion. The parameters that govern it are classified as key process parameters (Section 4).

Table 2.2: Step and cumulative product yield across the drug substance train.

Step	Unit operation	Step yield (%)	Cumulative yield (%)
3	Production Bioreactor	100.0	100.0
4	Harvest / Clarification	97.7	97.7
5	Protein A Chromatography	97.2	95.0

Step	Unit operation	Step yield (%)	Cumulative yield (%)
6	Low-pH Viral Inactivation	99.0	94.0
7	Cation Exchange Chromatography	95.2	89.5
8	Anion Exchange Chromatography	97.3	87.1
9	Small Virus Retentive Filtration	98.4	85.7
10	Ultrafiltration / Diafiltration (formulation)	97.1	83.2

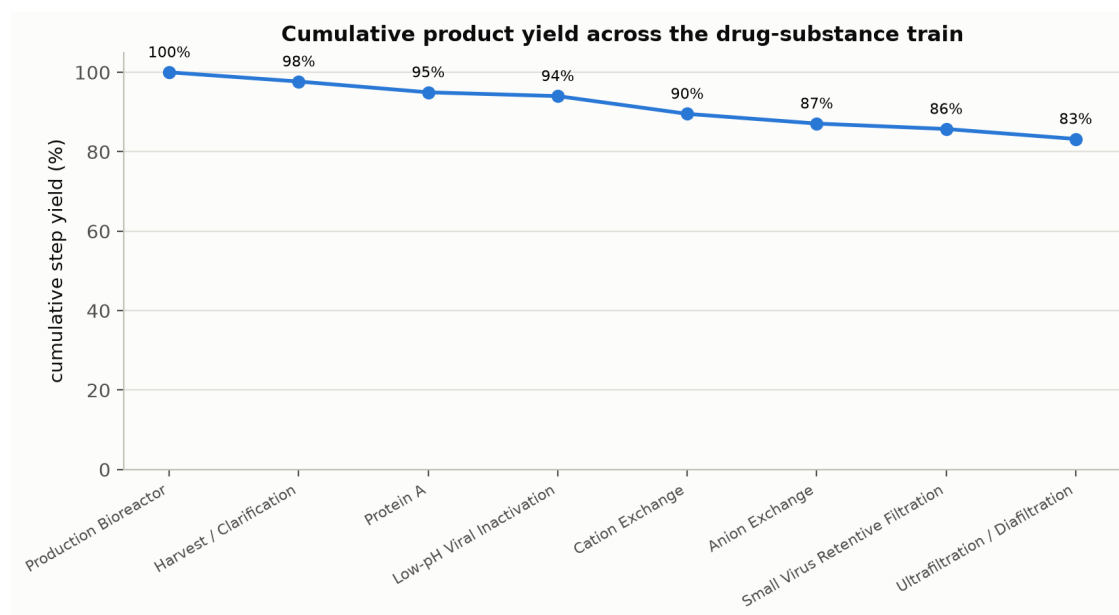


Figure 2.2: Cumulative product yield across the eight modelled unit operations.

The purification train reduces the impurity burden by several orders of magnitude. In the nominal batch, host cell protein leaves the capture step at 9,125 ng/mg, falls to 117 ng/mg across cation exchange and to 18.9 ng/mg across anion exchange. Leached Protein A enters the pool at the capture step at 3.63 ppm and is then cleared by the same two ion exchange steps. Aggregate rises across the low-pH hold, by 0.35 percentage points in the nominal batch, and is reduced 2.6-fold at cation exchange, which is why that step is described as the principal aggregate polish.

3 Consolidated quality attribute outcomes

3.1 The attributes and where they are set

The drug substance has 10 critical quality attributes, and every one of them met its acceptance criterion in the characterized process. Table 3.1 lists the register with the criticality of each attribute, the step that sets it and the report that characterizes it. Of the 10 attributes, 2 are cumulative viral clearance claims delivered by several steps together, and the remaining 8 are product quality or process impurity attributes measured on the drug substance itself.

Table 3.1: The drug substance quality attribute register, with the step that sets each attribute and the report that characterizes it.

CQA	Category	Critical-ity	Acceptance	Set by	Re- port
Afucosylation	glycosylation	H	2–13 % of glycans	Production Bioreactor	PCR-003
Galactosylation (total %Gal)	glycosylation	H	10–40 % (G1+G2 equiv.)	Production Bioreactor	PCR-003
High Mannose	glycosylation	VH	3–10 % of glycans	Production Bioreactor	PCR-003
Aggregates (HMW)	size variant	H	0–5 % HMW (SEC)	Production Bioreactor	PCR-003
Acidic charge variants (deamidation)	charge variant	VL	18–40 % (CEX/icIEF)	Production Bioreactor	PCR-003
Host Cell Protein (HCP)	process impurity	M-H	0–100 ng/mg	Production Bioreactor	PCR-003
Residual DNA	process impurity	VL	0–0.001 ng/dose	Production Bioreactor	PCR-003
Leached Protein A	process impurity	L-M	0–5 ppm	Protein A Chromatography	PCR-005
Viral clearance — XMuLV (enveloped)	viral safety	VH	16.7–30 log ₁₀ (cumulative)	Low-pH Viral Inactivation	PCR-006
Viral clearance — MVM (parvovirus)	viral safety	VH	8.6–20 log ₁₀ (cumulative)	Anion Exchange Chromatography	PCR-008

Setting an attribute and controlling it are different things, and for several attributes they happen at different steps. Host cell protein and residual DNA are generated in the culture and are cleared by Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008). Aggregate is formed in the bioreactor (PCR-003), rises across the low-pH hold (PCR-006) and is reduced at cation exchange (PCR-007), the principal polishing step for it. Leached Protein A is formed at the capture step (PCR-005) and is cleared by the two ion exchange steps. The 3 glycan attributes are formed inside the cell and in the culture fluid. The platform purification steps do not separate glycosylation variants of a monoclonal antibody (CMC Biotech Working Group 2009), so PCR-003 carries them alone. Acidic charge variants are also set in the culture, and the low-pH hold reports them as a response as well (PCR-006).

3.2 Outcome against the acceptance criteria

Table 3.2 gives the outcome of the campaign attribute by attribute, and each row compares the simulated commercial-scale distribution with the criterion that governs the attribute. For the impurity, size variant and charge variant attributes only the upper limit is an acceptance limit, so the criterion is one-sided and the capability index in Table 5.1 is one-sided with it. For the viral clearance attributes the criterion is a floor, and for the glycan attributes both limits apply. Across 2,000 simulated batches, 10 attributes met their criterion and none produced a value outside it.

Table 3.2: Simulated commercial-scale outcome for every drug substance quality attribute.

CQA	Criterion	Simulated range	Outcome
Afucosylation	2-13 % of glycans	6.265 – 9.967	met
Galactosylation (total %Gal)	10-40 % (G1+G2 equiv.)	20.94 – 32.22	met
High Mannose	3-10 % of glycans	4.284 – 7.978	met
Aggregates (HMW)	≤ 5 % HMW (SEC)	0.5011 – 1.15	met
Acidic charge variants (deamidation)	≤ 40 % (CEX/icIEF)	17.21 – 27.22	met
Host Cell Protein (HCP)	≤ 100 ng/mg	5.538 – 42.32	met
Residual DNA	≤ 0.001 ng/dose	0.0001 – 0.0003	met
Leached Protein A	≤ 5 ppm	0.0014 – 0.0056	met
Viral clearance — XMuLV (enveloped)	≥ 16.7 log10 (cumulative)	17.51 – 22.38	met
Viral clearance — MVM (parvovirus)	≥ 8.6 log10 (cumulative)	9.076 – 11.96	met

Two rows repay a second look. The simulated range for acidic charge variants extends below the lower figure shown for that attribute in Table 3.1, which is not an acceptance limit, because the specification for acidic charge variants is one-sided and the risk

it addresses is an elevated level. The second is leached Protein A, whose simulated concentrations lie far below the limit, so its capability index is correspondingly large and carries no information beyond that.

The margins are not uniform. High mannose and the two viral clearance attributes sit closest to their limits, and they are the attributes on which the operating region is effectively binding, with the capability figures given in Section 5. The impurity attributes clear their limits by a wide margin, which is a property of a train in which the capture step and both polishing steps remove them, and not a property of any single step.

The outcome in each row is a simulated one. The simulation varies the process parameters and propagates them through the fitted models of the scale-down studies, so it covers the variation those models describe and no other, and it does not cover raw material variability beyond what the models carry. It is not a measurement of commercial equipment, and the attribute levels are confirmed at scale during Stage 2.

4 Parameter classification summary

4.1 The classification across the train

Every parameter carried into the campaign has a classification, and the counts are 1 critical process parameter, 20 well-controlled critical process parameters, 10 key process parameters and 6 general process parameters. Figure 4.1 shows how they fall across the train. Table 4.1 lists the 21 quality-linked parameters with their set-points and normal operating ranges, and those ranges are the ranges the commercial process is asked to hold. The classification decisions are made in the report for each step, under SOP-4001.

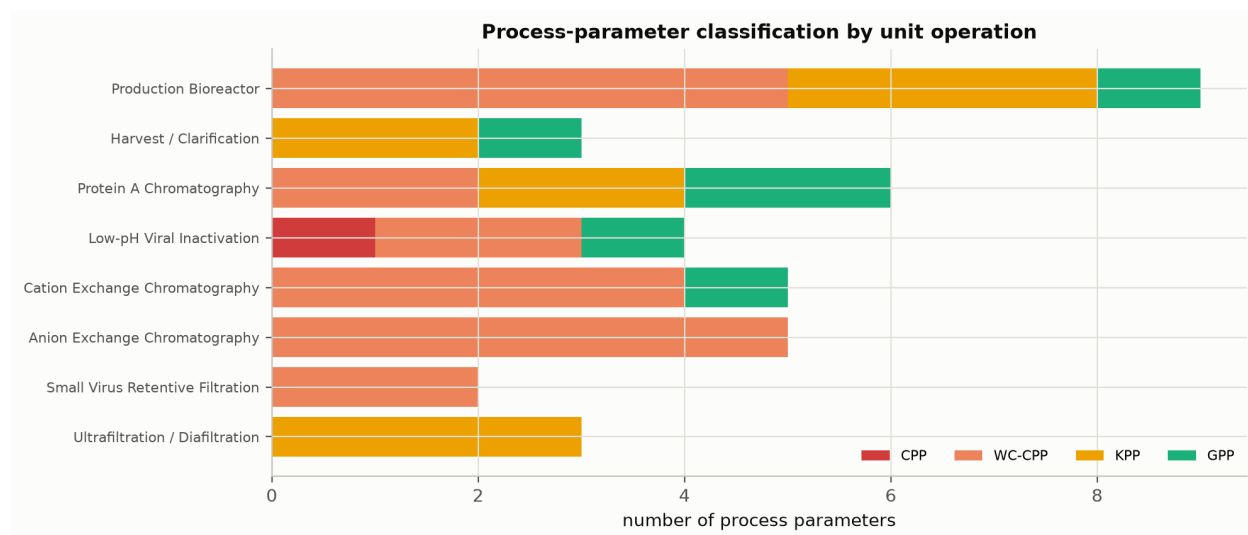


Figure 4.1: Post-characterization classification of process parameters by unit operation.

Table 4.1: The quality-linked process parameters of the drug substance train, with their set-points and normal operating ranges.

Unit operation	Parameter	Unit	Set-point	NOR	Class
Production Bioreactor	Culture pH	pH	6.85	6.75–6.95	WC-CPP
Production Bioreactor	Culture temperature	°C	35	34.5–35.5	WC-CPP
Production Bioreactor	Dissolved CO2 (pCO2)	mmHg	100	70–130	WC-CPP

Unit operation	Parameter	Unit	Set-point	NOR	Class
Production Bioreactor	Osmolality	mOsm	400	375-425	WC-CPP
Production Bioreactor	Culture duration	days	17	16-18	WC-CPP
Protein A Chromatography	Protein load	g/L resin	30	20-40	WC-CPP
Protein A Chromatography	Elution buffer pH	pH	3.55	3.4-3.7	WC-CPP
Low-pH Viral Inactivation	Inactivation pH	pH	3.5	3.4-3.6	CPP
Low-pH Viral Inactivation	Hold time	min	90	60-120	WC-CPP
Low-pH Viral Inactivation	Temperature	°C	21	19-23	WC-CPP
Cation Exchange Chromatography	Protein load	g/L resin	25	18-32	WC-CPP
Cation Exchange Chromatography	Load/Wash conductivity	mS/cm	5	4-6	WC-CPP
Cation Exchange Chromatography	Elution buffer pH	pH	6	5.85-6.15	WC-CPP
Cation Exchange Chromatography	Elution stop collect	OD desc.	1	0.7-1.3	WC-CPP
Anion Exchange Chromatography	Load pH	pH	7.5	7.35-7.65	WC-CPP
Anion Exchange Chromatography	Equil/Wash-1 conductivity	mS/cm	2.6	2.1-3.1	WC-CPP
Anion Exchange Chromatography	Load conductivity	mS/cm	5.5	4-7	WC-CPP
Anion Exchange Chromatography	Protein load	g/L resin	200	120-280	WC-CPP
Anion Exchange Chromatography	Operating flow rate	cm/hr	300	200-400	WC-CPP
Small Virus Retentive Filtration	Filtration volume (load)	L/m2	90	50-105	WC-CPP
Small Virus Retentive Filtration	Filtration pressure	psi	13	10-20	WC-CPP

The distinction between the two quality-linked classes is the risk of leaving the design space and not the size of the effect (CMC Biotech Working Group 2009). A well-controlled critical process parameter affects a critical quality attribute and has a low risk of moving outside the design space, whereas a critical process parameter carries a high risk of it. The class therefore follows from the control capability of the equipment as much as from the result of the study. The 20 well-controlled critical process parameters are spread across 6 unit operations, and every one of them is held within a normal operating range that lies at or inside the edges of the range

characterized for it. The characterized range is given parameter by parameter in the report for each step.

4.2 The exceptions

The process has 1 critical process parameter, the pH of the low-pH viral inactivation hold. Its normal operating range is 0.2 pH units wide, and the study characterized 0.8 pH units around it. The parameter governs the enveloped-virus clearance claim, an attribute of very high criticality (Table 3.1), which satisfies the first half of the definition. The second half is a judgement about how closely the step can be held to its range, and the assessment behind it is in PCR-006, which also records the classification. No other step needed the more stringent class.

The campaign classified 1 quality-linked parameter on a univariate assessment and not on a multivariate effect estimate. It is the operating flow rate of the anion exchange step, which RA-001 assigned to a univariate assessment and which PCR-008 then classified from the range given in Table 4.1, so its classification rests on a bounding argument and not on a measured multivariate effect.

Neither harvest and clarification nor the formulation step carries a quality-linked parameter (Figure 4.1). Neither step sets a critical quality attribute (Table 3.1), so their parameters are classified for process performance. The ranges they are held to protect yield, filter capacity and the concentration of the delivered pool.

4.3 Null results and what they are worth

A parameter that produced no measurable effect over the range studied is still classified, and the range over which nothing happened is recorded in the report for its step. On that basis 6 parameters are classified as general process parameters. The absence of an effect over a characterized range is evidence that the attribute is robust to the parameter over that range, but it is not evidence about conditions outside it. Each report states the range it covered.

5 Process capability

Commercial-scale capability was estimated for every drug substance quality attribute, and the lowest capability index in the set is 1.51. The estimate comes from simulating 2,000 batches, in which each batch draws every process parameter about its set-point, with a spread set by the width of its normal operating range and bounded by the range characterized, and propagates the drawn operating point through the fitted models of the scale-down studies. Capability is one-sided against the governing limit for the impurity, size variant and charge variant attributes and for the clearance attributes (Table 5.1), and two-sided for the glycan attributes.

Table 5.1: Commercial-scale process capability for every drug substance quality attribute.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Afucosylation	H	two_sided	2-13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10-40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3-10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0-5	0.753 ± 0.088	16.11
Acidic charge variants (deamidation)	VL	upper	18-40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0-0.001	0.0001 ± 0	10.81
Leached Protein A	L-M	upper	0-5	0.003 ± 0.0006	2785.30
Viral clearance — XMuLV (enveloped)	VH	lower	16.7-30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6-20	10.4 ± 0.4	1.51

Commercial-scale process capability (n = 2000 simulated batches)

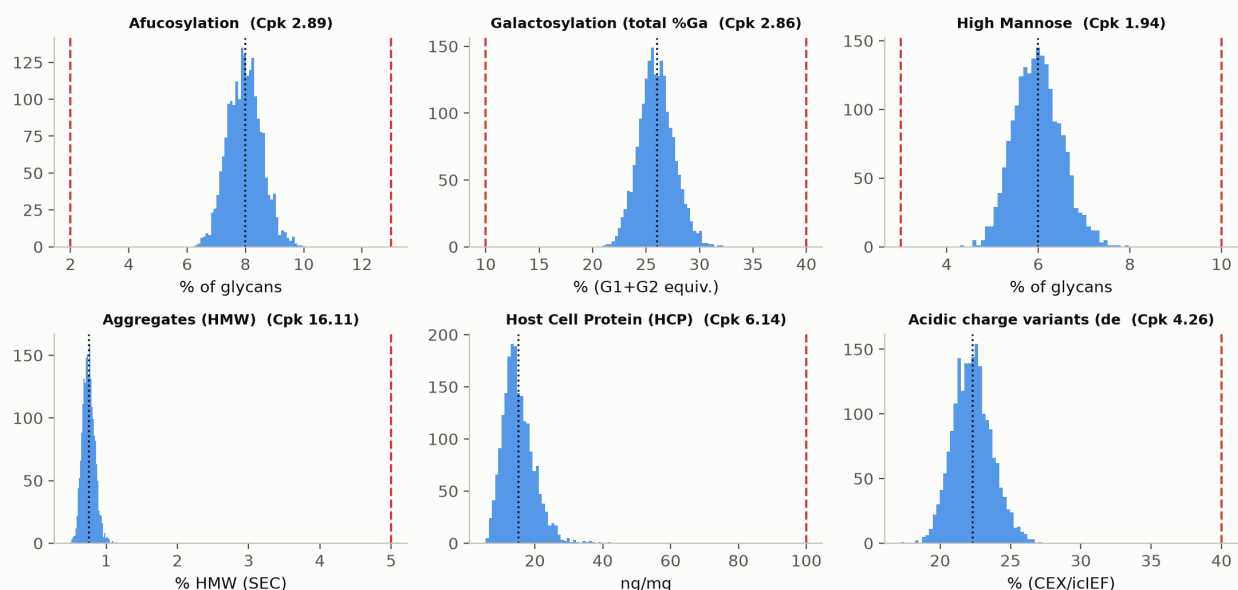


Figure 5.1: Simulated drug substance attribute distributions against their acceptance limits, annotated with the capability index.

The tightest capability in the process is cumulative MVM clearance, at 1.51, with a simulated mean 1.78 $\log_{10}$ above its floor, and cumulative XMuLV clearance is next, at 1.53 with a margin of 2.81 $\log_{10}$. Both are governed by more than one step, so neither figure is the property of a single unit operation, and the contributions are set out in Section 6. High mannose is the tightest of the attributes formed in the process, at 1.94, and it is governed by the bioreactor alone (PCR-003). The remaining attributes clear their limits by a wider margin, and the largest index in Table 5.1, for leached Protein A, reflects a pool concentration far below the limit and nothing else.

This report sets no acceptance threshold for the capability index, and each index is reported with the margin between the simulated mean and the governing limit, which is the quantity a control decision acts on.

The capability estimate is built on qualified scale-down models and inherits their assumptions about feed states and about scale independence. It describes the process operating near its set-points, so it says nothing about performance at the edges of the design space, which have not been run at commercial scale. It is not a measurement of the commercial process at all, and it is confirmed during Stage 2, when capability is re-estimated from process performance qualification batches and from the data of continued process verification.

6 Viral clearance summary

6.1 The modular claim

Cumulative viral clearance exceeds its requirement for both model viruses. Table 6.1 gives the credited log reduction of each contributing step, the cumulative total and the requirement, and Figure 6.1 shows the same figures against the requirement line. The claim is modular in the sense of ICH Q5A, which means that each step was spiked and studied independently on a qualified small-scale model and the step increments are then added (International Council for Harmonisation 2023a). The model viruses are XMuLV (an enveloped retrovirus) and MVM (a small non-enveloped parvovirus chosen as the worst case for physical removal).

Table 6.1: Modular viral clearance by credited step, in log10 reduction, against the drug substance requirement.

Step	XMuLV LRF	MVM LRF
Low-pH Viral Inactivation	7.10	0.00
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.87	10.03
Requirement (drug substance)	16.70	8.60

Cumulative viral clearance vs regulatory requirement

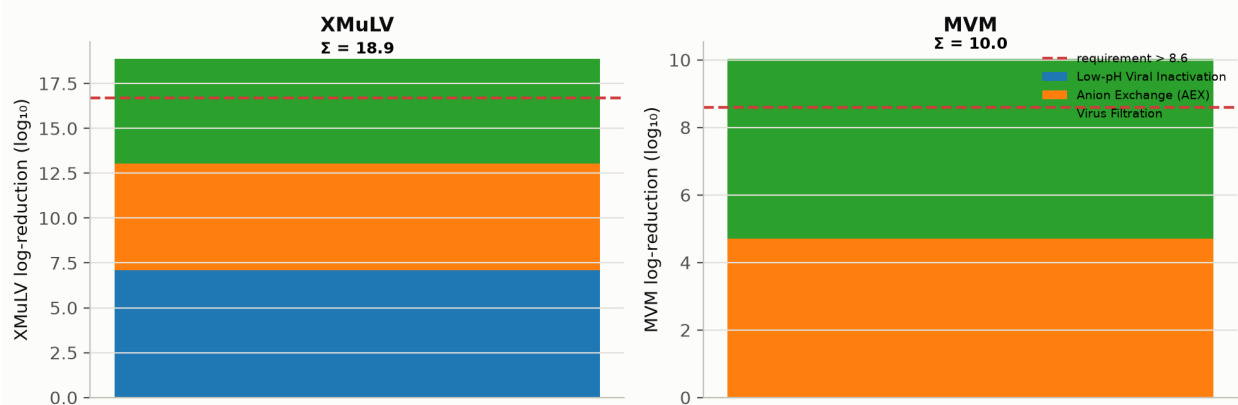


Figure 6.1: Modular viral clearance by credited step for both model viruses, against the requirement.

Table 6.1 credits 3 steps. The low-pH hold contributes $7.10 \log_{10}$ of XMuLV clearance and is the largest single contribution to the enveloped-virus claim (PCR-006), and it contributes nothing to the MVM claim, which Table 6.1 records as zero, because a parvovirus has no lipid envelope for low pH to act on. Anion exchange contributes $4.71 \log_{10}$ of MVM clearance in flow-through mode (PCR-008) and virus filtration contributes $5.32 \log_{10}$ (PCR-009), so the two together carry the small-virus claim, and the margins over the requirement are $2.17 \log_{10}$ for XMuLV and $1.43 \log_{10}$ for MVM.

6.2 Orthogonality, and what is not claimed

Adding the step increments is only defensible if the mechanisms are independent, and the three mechanisms differ. Low pH inactivates by disrupting the viral envelope, anion exchange partitions virus from product by charge in a step where the product flows through and the impurities bind, and virus filtration removes by size on a membrane rated for small non-enveloped viruses. A condition that reduces one of the three increments is therefore not expected to reduce the others. The campaign did not study conditions that act on more than one step at once, so that expectation rests on the mechanisms and not on a cross-step experiment.

No clearance is claimed for the Protein A step or for the cation exchange step, and neither contributes to the totals in Table 6.1. Neither step was studied for viral clearance in this campaign, so the cumulative claim is conservative to the extent of whatever those two steps deliver.

The log reduction values come from spiking studies on small-scale models and are not measurements of the commercial process. Each step's contribution holds over the ranges that step characterized, and the constraints are step specific, covering pH, hold time and temperature at the low-pH hold (PCR-006), load pH and conductivity at anion exchange (PCR-008), and volumetric load at virus filtration (PCR-009), where clearance falls as the load rises. The clearance attributes in Table 5.1 carry a simulated mean that differs from the nominal totals in Table 6.1, because the capability estimate varies the process parameters and the nominal batch does not.

7 Overall control strategy

7.1 What the control strategy has to do

The control strategy has two purposes. The first is to assure product quality and patient safety, and it does that by holding every quality-linked parameter inside the design space established by the characterization studies. The second is to assure that the commercial process runs consistently, and it does that by controlling the key process parameters and by monitoring the process attributes that report on the health of a batch. The two purposes need different evidence, which is why a key process parameter is controlled even though it was shown not to move a quality attribute (CMC Biotech Working Group 2009).

7.2 Parameter control

The quality-linked parameters and their normal operating ranges are given in Table 4.1, each is controlled to that range in routine manufacture, and each range sits inside the region the characterization studies covered. Movement inside the design space is not a regulatory change (International Council for Harmonisation 2009), but it is still assessed prospectively under the quality system before it is made (International Council for Harmonisation 2008), and movement outside the characterized region is a change and is handled through change control.

The critical process parameter, the pH of the low-pH hold, carries the tightest control requirement in the process, and its control is defined in PCR-006. The proven acceptable ranges that support each parameter's range are reported per step and per attribute in the report for that step, and are not reproduced here.

7.3 In-process control and monitoring

In-process controls are placed where an attribute can be measured early enough to act on it. Post-clarification turbidity is monitored at the harvest step and is one of the parameters that step controls (PCR-004). Pool host cell protein is measured after each chromatography step (PCR-005, PCR-007 and PCR-008), and pool aggregate after the low-pH hold and the cation exchange step, which is what makes the clearance chain described in Section 2 auditable batch by batch. The pH and the duration of the low-pH hold are recorded against the batch record, and the integrity test after virus

filtration is a release-relevant control on the small-virus claim. The concentration of the final pool is verified before the drug substance is dispensed.

7.4 Release testing

Release testing confirms on the batch what process control is designed to deliver. The 8 product quality and process impurity attributes in Table 3.2 are measured on the drug substance and compared with the criterion given there, so each of them is confirmed batch by batch once the process has run. The 2 viral clearance attributes are not confirmed in that way, because no test on the drug substance measures them. They are process claims established on spiked small-scale models (Section 6), and they are held by the operating ranges of the 3 credited steps and by the in-process controls above. A release result arrives after the batch has been made, so testing at release cannot by itself control an attribute (International Council for Harmonisation 2009). For this reason the control strategy places the burden on the parameter ranges and the in-process controls, and uses release testing to confirm the outcome.

7.5 Analytical control

Every attribute in Table 3.2 is measured by a validated method. Table 7.1 summarises the validated performance of 8 of the methods the campaign relies on, most of which measure a drug substance quality attribute. Turbidity and protein concentration are in that table for a different reason, because turbidity is an in-process control at the harvest step (PCR-004) and protein concentration governs the pool the formulation step delivers (PCR-010). Method performance matters to the control strategy for two reasons. It sets how much of the observed variation belongs to the process instead of to the measurement, and it sets how small a change the process can be held to.

Table 7.1: Validated performance of the analytical methods the campaign relies on.

Method	Title	Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3221	Aggregate by SEC-HPLC (verification-qualified)	3.8 %RSD	0.2 % HMW	99.1	0.34
AMV-3016	Leached Protein A by ELISA	6.5 %RSD	0.25 ppm	97.4	0.41
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3 log10 SD	1.5 log10 TCID50/mL	96	0.28
AMV-3019	Protein Concentration by A280 (UV)	1.2 %RSD	0.5 g/L	99.6	0.22
AMV-3010	Released N-Glycan Map (2-AB HILIC-UPLC)	2.5 %RSD	0.5 % glycan	99	0.3
AMV-3015	Turbidity by Nephelometry (NTU)	4 %RSD	0.5 NTU	98	0.38

Method	Title	Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32 log10 SD	1.4 log10 TCID50/mL	96	0.3
AMV-3012	Host-Cell Protein by ELISA	9.5 %RSD	1 ng/mg	95	0.4

The least precise method of those reported as a relative standard deviation is the host cell protein assay, at 9.5% RSD, and the largest share of observed variation attributable to a method is 0.41, for the leached Protein A assay. Neither limits the control strategy, because both attributes clear their limits by a wide margin (Table 3.2). The two infectivity assays are reported as a log10 standard deviation and not as a relative standard deviation, so their precision figures are not comparable with the rest of the table. Methods that do not appear in Table 7.1 carry their validated performance in their own method validation reports. Table 7.2 lists the controlled documents the unit operation studies cite from harvest onward, and the culture operation and analytical procedures the bioreactor study works to are cited in PCR-003.

Table 7.2: Controlled documents cited by the unit operation studies from harvest onward (Steps 4–10).

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP

Reference	Title	Type
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuvLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation

7.6 Life-cycle and feed-input controls

Three life-cycle controls sit behind the ranges in Table 4.1. Chromatography resins are managed for cycle life, packing and sanitization under SOP-2008. The clearance and polishing performance of a column is a function of its history as well as of its operating conditions. The virus retentive filter is integrity tested after use under SOP-2012, and the analytical methods transfer to the receiving site and are re-qualified there under the plan in PTP-001.

One feed-input control follows from a deviation. The first anion exchange execution used a load that had been held neutralized for 36 hours, which raised its acidic charge variant content well above the level of routine material (Section 8). The time the neutralized cation exchange pool is held before anion exchange loading is therefore treated as a control input for that step, and PCR-008 carries the assessment.

7.7 What the control strategy does not cover

The strategy described here covers the drug substance process from Step 3 to Step 10. It does not cover the seed train ahead of the production bioreactor, the cell bank, the raw material controls or the drug product process. All of those are described

elsewhere in the application. It does not by itself show that the commercial process is in a state of control, which is the purpose of Stage 2 and of the continued process verification that follows it.

8 Deviations across the campaign

The campaign recorded 17 deviations across the 8 unit operation studies, and one of them invalidated a designed experiment. Table 8.1 gives the register with the route by which each was detected and its disposition. The investigation, the root cause and the impact assessment for each are in the report named in the register, and are not repeated here.

Table 8.1: Deviations recorded across the characterization campaign.

De- via- tion	Re- port	Summary	Detected during	Disposition
DEV-003-01	PCR-003	Dissolved-CO2 probe drift during screening run 4	in-process monitoring vs offline reference	retained
DEV-003-02	PCR-003	Nutrient feed-1 under-delivery on a centre-point batch	post-execution mass reconciliation	retained
DEV-004-01	PCR-004	Depth-filter lot pre-use flush turbidity above alert	pre-use filter flush	retained
DEV-004-02	PCR-004	Post-clarification turbidity excursion above NOR on one run	in-process turbidity monitoring	retained
DEV-005-01	PCR-005	Elution-buffer pH prepared below target on an RSM run	buffer release testing	retained
DEV-005-02	PCR-005	Leached Protein A result initially out-of-trend; re-assayed	analytical review	retained
DEV-006-01	PCR-006	Hold-time record discrepancy between DCS and manual log	batch-record reconciliation	retained
DEV-006-02	PCR-006	XMuLV infectivity assay cytotoxicity at low dilution	assay review	retained
DEV-007-01	PCR-007	Expired Tris load/wash buffer lot used in screening centre-point run 17	post-execution documentation review	retained

De- via- tion	Re- port	Summary	Detected during	Disposition
DEV-007-02	PCR-007	Column inlet temperature excursion during RSM run 14	equipment history review	retained
DEV-008-01	PCR-008	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	invalidated and re-executed
DEV-008-02	PCR-008	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected by modelling and verification runs
DEV-008-03	PCR-008	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	retained
DEV-009-01	PCR-009	Transient filtration-pressure excursion above NOR	in-process pressure monitoring	retained
DEV-009-02	PCR-009	Filter membrane lot flux below vendor-typical value	pre-use flux verification	retained
DEV-010-01	PCR-010	Transmembrane-pressure excursion above NOR during concentration	in-process TMP monitoring	retained
DEV-010-02	PCR-010	Final DS concentration below target on one run; topped up	final concentration check	retained

8.1 The deviation that invalidated a study

The most serious deviation is DEV-008-01. The load used in the first anion exchange execution carried 33.5% acidic charge variants, against 21.0% in the requalified load, after the neutralized cation exchange eluate was held for 36 hours. A load in that condition is not representative of routine material, so the screening and response surface designs executed on it were invalidated for the purpose of defining an operating region, and both were re-executed in full on requalified load. The analysis reported in PCR-008 is the re-execution, and the first execution is retained there as a superseded dataset and referenced to confirm the root cause. No part of it is analysed in this report.

One further deviation at the same step was closed by analysis instead of by re-execution. The trailing-edge collection set-point used for the flow-through pool was more permissive than intended in both executions, at 180 mAU against a corrected 120 mAU.

The effect on the pool was modelled and then checked against 3 verification runs, and PCR-008 gives the assessment.

8.2 The retained deviations

In 15 cases the deviation was retained, which means the affected data were kept in the analysis with an argument that bounds the impact. Most fall into three kinds. Some are single-run excursions of a controlled parameter that stayed inside the range the study itself covered. The pressure excursion at virus filtration (DEV-009-01) and the transmembrane pressure excursion at the formulation step (DEV-010-01) are of that kind. Some are material events found before or after execution, such as the expired buffer lot at cation exchange (DEV-007-01) and the buffer released below its pH acceptance range at anion exchange (DEV-008-03). Others are analytical events resolved by re-test or by dilution, such as the cytotoxicity seen at low dilution in the XMuLV infectivity assay (DEV-006-02). The remainder are single events of other kinds. The dissolved carbon dioxide probe drift (DEV-003-01) and the feed under-delivery on a centre-point batch (DEV-003-02) are both bioreactor events. The others are the hold-time record discrepancy between the control system and the manual log at the low-pH hold (DEV-006-01) and the final pool concentration below target at the formulation step (DEV-010-02), which was corrected by topping up.

Most of the deviations in Table 8.1 were found by checks that run independently of the study analysis, among them buffer release testing, in-process monitoring, equipment history review, pre-use verification and post-execution reconciliation. The exceptions are analytical events. The leached Protein A result that fell out of trend at the capture step (DEV-005-02) and the assay cytotoxicity at the low-pH hold (DEV-006-02) were both found in the review of the study's own results. An instrumented scale-down system is involved in 3 of the deviations. Those systems are listed in Table 8.2 with their calibration status at the time the affected studies ran, and each was inside its calibration interval.

Table 8.2: Instrumented scale-down systems under calibration and change control.

Equip- ment	Description	Calibration status	Next calibration due
EQ-CHR-118	temperature-controlled chromatography chamber	in calibration at execution	2026-05-30
EQ-BRX-205	production scale-down bioreactor pCO ₂ probe	in calibration at execution	2026-06-14
EQ-TFF-142	ultrafiltration / diafiltration TFF skid	in calibration at execution	2026-04-22

9 Conclusions and Stage 2 readiness

The A-Mab drug substance process is characterized across Steps 3-10 and the campaign supports a defined control strategy. Every one of the 37 process parameters has a classification, 21 of them are quality-linked and are controlled to the ranges in Table 4.1, and the remaining 16 are controlled for process consistency. Each unit operation has a plan, a report and an operating region. At the 6 steps that carried a designed experiment the operating region is derived from that step's response surface models, and at harvest and the formulation step it rests on the univariate assessments those reports present.

Every drug substance quality attribute met its acceptance criterion in the simulated commercial process. The lowest capability index is 1.51, for cumulative MVM clearance, and the tightest of the attributes the process forms is high mannose at 1.94. Cumulative viral clearance exceeds the requirement by $2.17 \log_{10}$ for XMuLV and by $1.43 \log_{10}$ for MVM, on 3 steps with independent mechanisms. The campaign recorded 17 deviations, one of which forced a study to be re-executed in full, and each was dispositioned in the report for its step.

The capability and clearance figures are estimated on qualified scale-down models and are not measured on commercial equipment, and they describe the process operating near its set-points, so they say nothing about the edges of the design space, which have not been run at scale. The viral clearance values come from small-scale spiking studies. Nothing here rests on a screening model, and no proven acceptable range extends beyond the range its step characterized.

What remains is Stage 2. The process transfers to the receiving site under PTP-001, process performance qualification batches are executed there, capability is re-estimated from those batches instead of from simulation, and the analytical methods transfer and are re-qualified at the receiving site. Continued process verification then runs for the life of the product. This report does not qualify the commercial process, and it makes no claim about the performance of any batch made at commercial scale.

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